

Research Interests

Human-Computer Interaction, Machine Learning for Human Sensing, On-Device AI

Education

New York University, Tandon School of Engineering September 2026 - Present
Ph.D. Student in Human-Centered Technology, Innovation & Design, advised by Prof. Vedant Das Swain
Brooklyn, NY

Seoul National University March 2017 - February 2024
Bachelor of Science in Computer Science and Engineering
Seoul, South Korea

Overall GPA: 3.53/4.00, Graduated *cum laude*

Research Experience

Post-Baccalaureate Research Fellow September 2024 - June 2026
Northwestern University, SPICE Lab, advised by Prof. Karan Ahuja
Evanston, IL

- Contributed to 5 research publications spanning on-device ML, human activity sensing, and real-time XR avatars.
- Presented research demonstrations to industry partners: **Google Platform and Experiences, Apple, Qualcomm XR, Adobe Research, Dolby Studios.**

Professional Experience

Full Stack Game Developer (Alternative Military Service) September 2019 - February 2023
Awesomepiece
Seoul, South Korea

CUDA Engineer Intern July 2019 - August 2019
MindsLab
Pangyo, South Korea

Publications

- Chenfeng Gao*, **Taeyoung Yeon***, Sungheon Park, Vasco Xu, Anish Prabhu, Mar Gonzalez-Franco, Karan Ahuja. **FlowAvatar: Real-Time Full-Body Avatars from Sparse Egocentric Inputs on Consumer XR Devices.** *To appear in Proceedings of the 2026 IEEE International Symposium on Mixed and Augmented Reality (ISMAR '26).* 2026.
- Yiquan Li*, **Taeyoung Yeon***, Chenfeng Gao, Vasco Xu, Xuanyou Liu, Karan Ahuja. **MARIO: Motion-Augmented Real-Time Multi-Sensor Inertial Odometry.** *Proceedings of the 2026 IEEE/CVF Conference on Computer Vision and Pattern Recognition, Findings (CVPR '26).* 2026.
- Taeyoung Yeon**, Vasco Xu, Hank Hoffman, Karan Ahuja. **WatchHAR: Real-time On-device Human Activity Recognition System for Smartwatches.** *Proceedings of the 27th ACM International Conference on Multimodal Interaction (ICMI '25).* 2025.

Technical Skills

Programming Languages: Python, Java, C#, JavaScript, Kotlin, Swift, CUDA

Machine Learning: TensorFlow, PyTorch, CoreML, scikit-learn, model optimization

Mobile & VR Development: Unity (Meta SDK), iOS/Android development, cross-platform frameworks

Backend & Infrastructure: Spring Boot, Netty, Django, React, MySQL, Google Cloud Platform

Academic Services

Reviewer: ICMI 2026, UbiComp/IMWUT 2026 (*2 Outstanding Reviewer Recognitions*)

References

Prof. Vedant Das Swain

Assistant Professor, Department of Technology Management and Innovation, NYU Tandon School of Engineering
v.das.swain@nyu.edu

Prof. Karan Ahuja

Lisa Wissner-Slivka & Benjamin Slivka Assistant Professor, Department of Computer Science, Northwestern University
kahuja@northwestern.edu